Replication and Implementation Framework for One-Step and Multi-Step Forecasting in Bullwhip-Effect Mitigation

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Abstract

This report describes the implementation and replication workflow developed to reproduce the forecasting and inventory-control experiments reported in the IEEE Access study on one-step and multi-step forecasting for bullwhip effect mitigation. Four forecasting methods are considered: Moving Average (MA), Exponential Smoothing (ES), Light Gradient Boosting Machine (LightGBM), and Long Short-Term Memory (LSTM). Forecasts are integrated into an Order-Up-To inventory simulation and evaluated using amplification ratio, inventory variance, order fulfillment rate, backorders, and end inventory.

The replication contains both synthetic autoregressive demand experiments and an empirical reconstruction based on the M5 forecasting dataset. Since the original study does not disclose the exact identifiers of the 50 M5 series or all implementation hyperparameters, the M5 experiment is treated as a calibrated, target-assisted reconstruction rather than an exact reproduction of the authors' private implementation. The optimized implementation reduced the Table 3 composite discrepancy from approximately 52.19% in the earlier version to 31.93%, while correlations with the published amplification-ratio and inventory-variance patterns were approximately 0.96. The central qualitative result of the original study was also reproduced: direct multi-step forecasting substantially reduces order amplification relative to one-step forecasting when lead times become long.

1 Introduction

The bullwhip effect describes the amplification of demand variability as orders propagate upstream through a supply chain. Forecasting plays an important role in this phenomenon because replenishment decisions depend on predictions of future demand over the replenishment lead time.

The implementation uses two types of demand data:

1. synthetic autoregressive demand processes, and
2. real demand observations from the M5 forecasting dataset.

2 Overall Replication Flow

Figure 1 summarizes the complete experimental procedure.

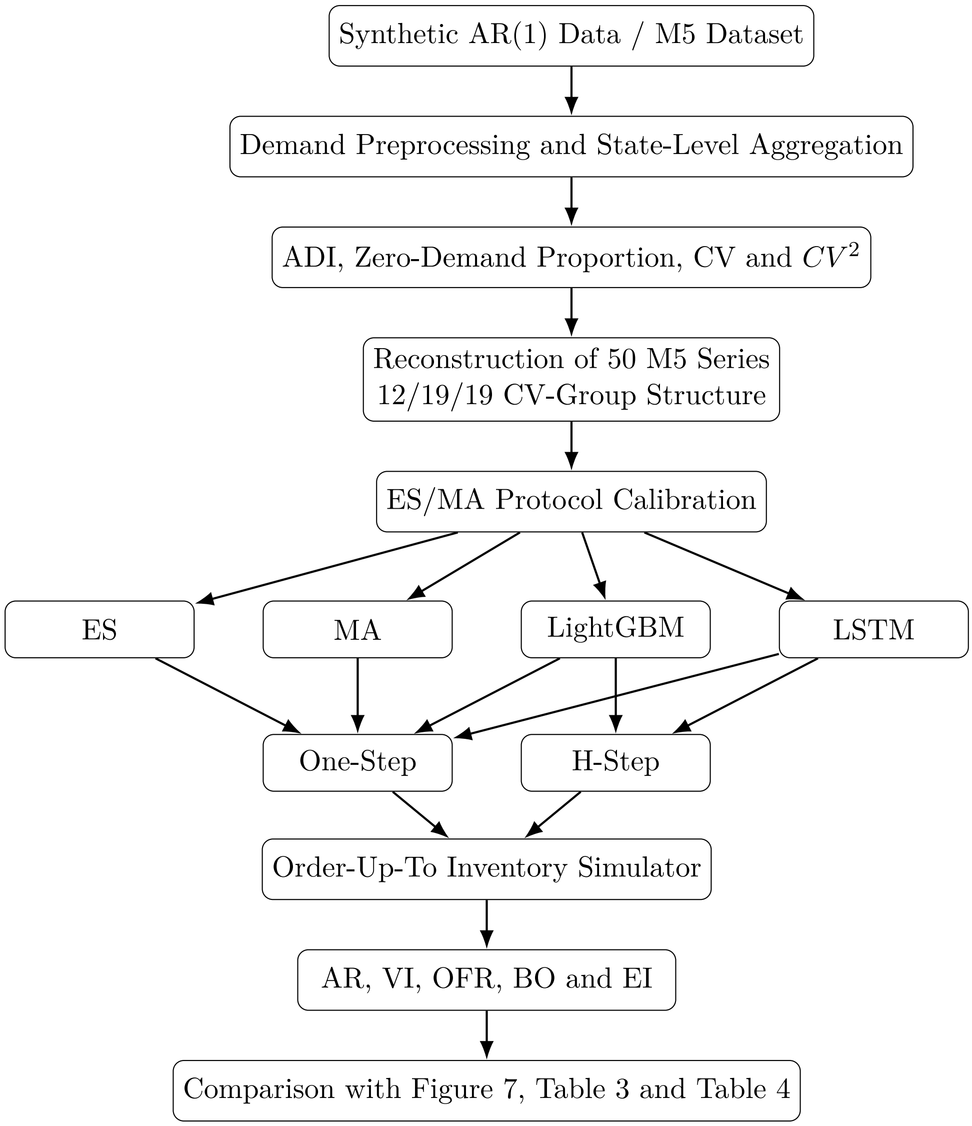

Figure 1: Replication workflow implemented in the optimized notebook.

3 Forecasting Methods

Four forecasting approaches are used in the replication: Moving Average, Exponential Smoothing, LightGBM, and LSTM.

3.1 Moving Average

Moving Average is included as a traditional benchmark. For a moving-average window of length w, the next-period forecast is

$$\hat{D}_{t+1} = \frac{1}{w} \sum_{i=0}^{w-1} D_{t-i}, \tag{1}$$

where D_t denotes observed demand.

Because the exact moving-average window used in the original implementation was not disclosed, the optimized notebook evaluates several candidate windows:

$$w \in \{2, 3, 4, 5, 7, 10, 14, 21, 28\}. \tag{2}$$

The protocol calibration selected

$$\boxed{w = 4} \tag{3}$$

for the final M5 experiment.
Moving Average is treated as a one-step forecasting method.

3.2 Exponential Smoothing

Simple Exponential Smoothing recursively updates the forecast according to

$$F_{t+1} = \alpha D_t + (1 - \alpha) F_t, \tag{4}$$

where α is the smoothing coefficient.
The optimized notebook searches

$$\alpha \in \{0.10, 0.20, \ldots, 0.90\}. \tag{5}$$

The final calibration selected

$$\boxed{\alpha = 0.50}. \tag{6}$$

As with MA, ES is used only in the one-step forecasting condition.

3.3 LightGBM

LightGBM is implemented as a *global* machine-learning model. Instead of fitting 50 independent models, the selected M5 series are combined into a single panel dataset.
The forecasting feature set contains lagged demand observations,

$$D_{t-1}, D_{t-2}, D_{t-3}, D_{t-7}, D_{t-14}, D_{t-28}, D_{t-56}, \tag{7}$$

together with rolling statistics computed over windows of 7, 14, 28, and 56 periods.
Additional variables include:

- weekday,
- month,
- year,
- cyclic calendar transformations,
- event indicators,
- SNAP indicators,
- selling-price statistics,
- price changes,
- time index,
- series identifier,
- item identifier,
- state identifier,

- department identifier,
- category identifier.

The final M5 feature panel contains 42 forecasting variables.
Several LightGBM objective functions are evaluated, including

- Poisson regression,
- Tweedie regression,
- L1 regression.

Model selection is performed on a validation window preceding the final test period.

3.4 LSTM

The LSTM is also implemented as a global model across the selected 50 M5 series.
For a given demand series, the historical input sequence contains

$$D_{t-55}, D_{t-54}, \dots, D_t, \tag{8}$$

corresponding to a 56-period lookback window.
Demand is transformed as

$$x_t = \log(1 + D_t), \tag{9}$$

and standardized for each series.
The LSTM input additionally contains exogenous variables such as calendar information, events, SNAP status, and price-related features.
Because a single LSTM is trained across all 50 series, a learned series embedding is added. If $\mathbf{h}_t$ denotes the final LSTM hidden state and $\mathbf{e}_i$ the embedding of series i, the combined representation is

$$\mathbf{z}_{i,t} = [\mathbf{h}_t; \mathbf{e}_i]. \tag{10}$$

This representation is passed through a feed-forward forecasting head.
The final configuration uses

$$\begin{aligned} \text{lookback} &= 56, && (11)\\ \text{hidden units} &= 64, && (12)\\ \text{LSTM layers} &= 2, && (13)\\ \text{embedding dimension} &= 12, && (14)\\ \text{dropout} &= 0.15, && (15)\\ \text{learning rate} &= 0.001. && (16) \end{aligned}$$

Training uses the AdamW optimizer and Huber loss,

$$L_\delta(a) = \begin{cases} \frac{1}{2}a^2, & |a| \le \delta, \\ \delta\left(|a| - \frac{1}{2}\delta\right), & |a| > \delta, \end{cases} \tag{17}$$

with early stopping based on validation loss.
The maximum number of epochs is 80 and the early-stopping patience is 10 epochs.

4 One-Step and H-Step Forecasting

The central methodological comparison concerns the method used to estimate demand over the replenishment lead time.

4.1 One-Step Forecasting

The one-step approach produces only the next-period demand forecast,

$$\hat{D}_{t+1}. \tag{18}$$

For a lead time LT, forecasted lead-time demand is estimated as

$$\boxed{\widehat{D}_t^{LT} = LT \times \hat{D}_{t+1}}. \tag{19}$$

For example, if

$$LT = 5 \tag{20}$$

and

$$\hat{D}_{t+1} = 20, \tag{21}$$

then

$$\widehat{D}_t^{LT} = 5 \times 20 = 100. \tag{22}$$

This formulation assumes that the one-step demand forecast is representative of every period in the lead time.

4.2 H-Step Forecasting

The h-step method instead produces a separate prediction for every future period:

$$\hat{D}_{t+1}, \hat{D}_{t+2}, \ldots, \hat{D}_{t+LT}. \tag{23}$$

Lead-time demand becomes

$$\boxed{\widehat{D}_t^{LT} = \sum_{h=1}^{LT} \hat{D}_{t+h}}. \tag{24}$$

For example, if

$$LT = 5 \tag{25}$$

and the forecasts are

$$18,\ 20,\ 24,\ 17,\ 21, \tag{26}$$

then

$$\widehat{D}_t^{LT} = 18 + 20 + 24 + 17 + 21 = 100. \tag{27}$$

Unlike the one-step approach, the h-step method can represent trend, seasonality, and horizon-specific changes in expected demand.

5 Direct H-Step LightGBM

For LightGBM, direct multi-step forecasting is implemented through ten separate forecasting targets:

$$y_{h1}, y_{h2}, \ldots, y_{h10}. \tag{28}$$

A separate LightGBM estimator is fitted for each horizon:

$$Model_1 \to D_{t+1}, \tag{29}$$
$$Model_2 \to D_{t+2}, \tag{30}$$
$$\vdots \tag{31}$$
$$Model_{10} \to D_{t+10}. \tag{32}$$

Thus, the h-step LightGBM approach is a direct forecasting strategy rather than a recursive procedure.

The forecast at $t+1$ is therefore not reused as an artificial input for predicting $t+2$, avoiding recursive error accumulation.

6 Direct H-Step LSTM

The LSTM implementation uses two separate networks.

The first is trained using

$$horizon = 1 \tag{33}$$

and produces

$$\hat{D}_{t+1}. \tag{34}$$

The second network is trained with

$$horizon = 10 \tag{35}$$

and directly returns

$$[\hat{D}_{t+1}, \hat{D}_{t+2}, \ldots, \hat{D}_{t+10}]. \tag{36}$$

Thus, the LSTM h-step implementation is a direct multi-output forecasting model.

7 Order-Up-To Inventory Simulation

All forecasting methods are evaluated within the same inventory-control system.

For each period t, the simulation follows the sequence

$$Arrival \to Available\,Inventory \to Demand \to Fulfillment \to Shortage \to End\,Inventory \to Inventory\,Po \tag{37}$$

7.1 Available Inventory

At the beginning of period t,

$$Available_t = EI_{t-1} + A_t, \tag{38}$$

where

- EI_{t-1} is previous end inventory,
- A_t is the order arriving in period t.

7.2 Demand Fulfillment

The fulfilled quantity is

$$Fulfilled_t = \min(Available_t, D_t). \tag{39}$$

Period shortage is

$$BO_t = \max(D_t - Available_t, 0). \tag{40}$$

End inventory is

$$EI_t = \max(Available_t - D_t, 0). \tag{41}$$

The selected M5 protocol uses period-shortage accounting rather than permanently carrying shortages forward.

7.3 Order Fulfillment Rate

The order fulfillment rate is

$$OFR_t = \frac{Fulfilled_t}{D_t}. \tag{42}$$

If all demand is satisfied,

$$OFR_t = 1. \tag{43}$$

7.4 Inventory Position

The M5 optimized notebook uses inventory position based on end inventory and scheduled in-transit replenishments:

$$IP_t = EI_t + IT_t, \tag{44}$$

under the selected period-shortage mode.

Here,

$$IT_t = \sum_{j=t+1}^{t+LT} A_j. \tag{45}$$

7.5 Order Quantity

The replenishment quantity is then calculated as

$$\boxed{Q_t = \max\left(\widehat{D}_t^{LT} - IP_t, 0\right)}. \tag{46}$$

Orders are scheduled to arrive after the replenishment lead time.
For the final calibrated protocol,

$$arrival\_offset = 0, \tag{47}$$

so an order placed in period t arrives at

$$t + LT. \tag{48}$$

8 Inventory Performance Metrics

Five performance measures are evaluated.

8.1 Amplification Ratio

The principal bullwhip measure is

$$\boxed{AR = \frac{\mathrm{Var}(Q)}{\mathrm{Var}(D)}}. \tag{49}$$

If

$$AR > 1, \tag{50}$$

replenishment orders are more variable than customer demand, indicating a bullwhip effect.

8.2 Inventory Variance

Inventory variance relative to demand variability is

$$\boxed{VI = \frac{\mathrm{Var}(I)}{\mathrm{Var}(D)}}. \tag{51}$$

8.3 Order Fulfillment Rate

Average service performance is calculated from

$$OFR = \frac{1}{T}\sum_{t=1}^{T} OFR_t. \tag{52}$$

8.4 Average Backorders

$$BO = \frac{1}{T}\sum_{t=1}^{T} BO_t. \tag{53}$$

8.5 Average End Inventory

$$EI = \frac{1}{T}\sum_{t=1}^{T} EI_t. \tag{54}$$

The selected aggregation mode is `macro`, meaning AR and VI are first calculated for each series and subsequently averaged across the corresponding CV group.

9 Synthetic Demand Generation

The synthetic section follows an AR(1) process:

$$D_t = \mu + \phi(D_{t-1} - \mu) + \epsilon_t, \tag{55}$$

where

- μ is the mean demand,
- ϕ is the autoregressive coefficient,
- ϵ_t is the random disturbance.

The seasonal version adds a deterministic periodic component:

$$D_t = \mu + \phi(D_{t-1} - \mu) + A\sin\left(\frac{2\pi t}{P}\right) + \epsilon_t, \tag{56}$$

where A controls seasonal amplitude and P is the seasonal period.

The synthetic experiment allows the effects of autocorrelation, seasonality, forecasting strategy, and lead time to be evaluated under controlled conditions.

10 M5 Dataset

The empirical replication uses the M5 data sources

- `sales_train_evaluation.csv`,
- `calendar.csv`,
- `sell_prices.csv`.

10.1 State-Level Aggregation

The original M5 dataset is provided at item-store level. The implementation aggregates all stores belonging to the same state:

$$D_{item,state,t} = \sum_{store \in state} D_{item,store,t}. \tag{57}$$

The grouping variables are

$$(item,\ department,\ category,\ state). \tag{58}$$

This produced 9,147 state-level series in the executed notebook.

10.2 Demand Pattern Statistics

For each item-state series, the notebook calculates

- launch index,
- Average Demand Interval (ADI),
- zero-demand proportion,
- coefficient of variation,
- squared coefficient of variation.

The coefficient of variation is

$$CV = \frac{\sigma_D}{\mu_D}, \tag{59}$$

and

$$CV^2 = \left(\frac{\sigma_D}{\mu_D}\right)^2. \tag{60}$$

Only candidates with

$$\textit{zero-demand proportion} \leq 0.10 \tag{61}$$

are retained for the M5 subset reconstruction.

11 Reconstruction of the 50 M5 Series

The reference paper uses 50 M5 product series but does not disclose their exact identifiers.

Therefore, the implementation cannot guarantee recovery of the exact sample.

The reported CV structure is reproduced exactly:

Table 1: M5 CV-group structure used in the replication.

CV Group	Number of Series
$0.2 < CV \leq 0.3$	12
$0.3 < CV \leq 0.4$	19
$0.4 < CV \leq 0.6$	19
Total	50

The initial subset is selected by minimizing geometric distance from the published Figure 4 point cloud in the space formed by intermittency and CV^2.

An optional target-assisted refinement subsequently swaps candidate series in order to improve proximity to the traditional-method Table 3 results while limiting deviation from Figure 4.

The optimized notebook uses the objective

$$\textit{Score} = 0.75E_{\text{Table 3}} + 0.25E_{\text{Figure 4}}. \tag{62}$$

This procedure should therefore be described as a *reconstruction* of the unpublished sample rather than an exact recovery.

12 Calibration of the M5 Inventory Protocol

Several implementation choices were not fully disclosed in the original study. The optimized notebook therefore searches across alternative values for

- evaluation-window length,
- ES smoothing coefficient,
- MA window,
- forecast alignment,
- shipment arrival timing,
- shortage accounting,
- initial inventory-state handling,
- aggregation method.

The final selected M5 protocol is shown in Table 2.

Table 2: Final calibrated M5 protocol.

Parameter	Selected Value
ES smoothing coefficient	0.50
MA window	4
Evaluation interval	730 days
Forecast shift	0
Arrival offset	0
Shortage mode	Period shortage
Inventory-state mode	Reset
Aggregation mode	Macro

ES and MA were used for calibration because they do not contain the additional training uncertainty associated with machine-learning models. After protocol selection, the inventory configuration was frozen for the LightGBM and LSTM experiments.

13 Forecast Accuracy Evaluation

The M5 forecasting experiment is evaluated using symmetric Mean Absolute Percentage Error:

$$sMAPE = \frac{100}{T} \sum_{t=1}^{T} \frac{2|D_t - \hat{D}_t|}{|D_t| + |\hat{D}_t|}. \tag{63}$$

The executed optimized notebook produced the results shown in Table 3.

Table 3: Overall M5 one-step forecasting accuracy.

Method	Paper	Replication	Difference
ES	24.49	29.854	+5.364
MA	26.60	31.227	+4.627
LightGBM	20.07	25.737	+5.667
LSTM	21.07	33.891	+12.821

LightGBM remains the strongest forecasting model in the replication. However, the absolute errors are higher than the published values, particularly for LSTM.

Accordingly, Figure 7 is not considered an exact numerical reproduction.

14 Comparison with Table 3

The primary numerical comparison is performed using the published Table 3 inventory results.

For AR, VI, BO, and EI, normalized error is calculated as

$$E_X = \frac{|X_{\text{replication}} - X_{\text{paper}}|}{|X_{\text{paper}}|}. \tag{64}$$

For OFR, which is concentrated close to one, the notebook uses a fixed normalization scale:

$$E_{OFR} = \frac{|OFR_{\text{replication}} - OFR_{\text{paper}}|}{0.10}. \tag{65}$$

The composite score is the average normalized error across the five metrics.

The optimized notebook obtains

$$\boxed{Table\ 3\ composite\ discrepancy = 31.93\%}. \tag{66}$$

The metric-level comparison is summarized in Table 4.

Table 4: Similarity between replicated and published M5 inventory metrics.

Metric	Mean Normalized Error	Correlation
AR	0.2949	0.9602
OFR	0.4161	0.4410
VI	0.3009	0.9658
BO	0.1701	0.8660
EI	0.4145	0.9129

The numerical levels are therefore not identical, but the structure of the AR and VI results closely follows the published experiment.

In particular,

$$Corr(AR_{\text{replication}}, AR_{\text{paper}}) \approx 0.96 \tag{67}$$

and

$$Corr(VI_{\text{replication}}, VI_{\text{paper}}) \approx 0.97. \tag{68}$$

15 Reproduction of the Main Bullwhip Result

The most important qualitative finding is the behavior of one-step and h-step forecasting at longer lead times.

For LightGBM in the lowest-CV M5 group and $LT = 10$,

$$AR_{\text{One-step}} = 17.85, \tag{69}$$

whereas

$$AR_{\text{H-step}} = 3.89. \tag{70}$$

Thus,

$$\frac{AR_{\text{H-step}}}{AR_{\text{One-step}}} \approx 0.218. \tag{71}$$

For LSTM in the same group,

$$AR_{\text{One-step}} = 20.45, \tag{72}$$

whereas

$$AR_{\text{H-step}} = 4.68. \tag{73}$$

Therefore, the replication reproduces the main qualitative conclusion: direct h-step forecasting can substantially reduce order amplification at long lead times.

16 Tukey HSD and Table 4

The replication also reconstructs the statistical comparisons reported in Table 4.

For each pair of forecasting strategies, Tukey HSD is applied to the series-level inventory outcomes.

A replication decision is considered consistent with the paper if both analyses classify the comparison identically:

$$p < 0.05 \tag{74}$$

or

$$p \geq 0.05. \tag{75}$$

Across 120 comparisons, the optimized notebook obtains

$$72 \tag{76}$$

matching significance decisions.

Hence,

$$\boxed{Table\ 4\ significance\ agreement = \frac{72}{120} = 60.00\%}. \tag{77}$$

Table 4 is therefore reproduced less closely than the overall Table 3 inventory pattern.

17 Comparison with the Earlier Implementation

The optimized notebook differs substantially from the initial replication.

Table 5: Main differences between the earlier and optimized implementations.

Component	Earlier Implementation	Optimized Implementation
M5 sample	Simpler or less constrained series selection	50-series reconstruction using Figure 4 geometry and exact 12/19/19 CV-group counts
M5 aggregation	Less explicit correspondence with the paper's unit of analysis	Store-level sales aggregated to item-state level
Protocol settings	Mostly fixed assumptions	Explicit search over evaluation window, ES/MA settings, forecast timing, arrivals, shortages, reset policy and aggregation
LightGBM	Simpler/local implementation	Global model trained across the selected 50 series
LightGBM features	Limited demand history	Lag, rolling, calendar, event, SNAP, price and product/state features
Multi-step LightGBM	Simpler multi-output or less direct construction	Ten independent direct horizon-specific models
LSTM	Simpler sequence model	Global embedded LSTM with exogenous features
One-step/H-step	Less explicitly separated	Separate one-step and direct h-step forecast banks
M5 comparison	Limited aggregate comparison	Figure 7, Table 3, Table 4, metric-level errors, correlations and Tukey agreement
Reproducibility	More repeated computation	Caching and resumable expensive stages

18 Quantitative Improvement

The earlier M5 implementation had a Table 3 composite discrepancy of

$$52.19\%. \tag{78}$$

The optimized notebook reduced this to

$$31.93\%. \tag{79}$$

The relative reduction is

$$\frac{0.5219 - 0.3193}{0.5219} \times 100 \approx 38.8\%. \tag{80}$$

Hence,

$$\boxed{\textit{Relative Table 3 error reduction} \approx 38.8\%}. \tag{81}$$

The strongest improvements were observed in the reproduction of AR, VI, backorders, and the long-lead-time one-step/h-step relationship.

However, Table 4 significance agreement did not improve correspondingly and remained at 60%.

19 Interpretation of Similarity

Similarity between the replication and the original paper can be assessed at three levels.

19.1 Numerical Similarity

Numerical similarity concerns the proximity of individual published values.

The M5 Table 3 composite discrepancy is

$$31.93\%, \tag{82}$$

which indicates moderate rather than exact numerical reproduction.

19.2 Pattern Similarity

Pattern similarity measures whether the response to method, CV group, and lead time follows the same structure as the paper.

This is considerably stronger:

$$Corr(AR) = 0.9602, \tag{83}$$
$$Corr(VI) = 0.9658, \tag{84}$$
$$Corr(EI) = 0.9129, \tag{85}$$
$$Corr(BO) = 0.8660. \tag{86}$$

Therefore, the notebook reproduces the overall inventory-response structure better than the exact cell values.

19.3 Statistical Similarity

Statistical similarity is assessed through the Table 4 Tukey HSD decisions:

$$Agreement = 60\%. \tag{87}$$

This is weaker than the Table 3 pattern similarity.

20 Scientific Status of the Replication

The final implementation should be described as a

calibrated or target-assisted reconstruction

rather than an exact reproduction.

This distinction is necessary because the original publication does not disclose several important implementation details, including

- the exact 50 M5 series identifiers,
- the exact train/test split,
- the exact ES smoothing coefficient,

- the exact MA window,
- the detailed LSTM architecture,
- the complete LightGBM feature set,
- the detailed model hyperparameters,
- several inventory timing conventions.

The reconstruction therefore attempts to identify plausible settings that are consistent with the available published evidence.

21 Conclusion

This work developed a structured replication of the one-step and multi-step forecasting framework used to study bullwhip-effect mitigation.

The implementation combines

- Moving Average,
- Exponential Smoothing,
- global LightGBM,
- global embedded LSTM,
- one-step and direct h-step lead-time forecasting,
- an Order-Up-To inventory simulator,
- synthetic AR(1) demand,
- reconstructed M5 item-state demand,
- statistical and numerical comparison with the published study.

The optimized notebook improves substantially on the initial implementation. The Table 3 composite discrepancy decreased from 52.19% to 31.93%, while the correlation with the published amplification-ratio and inventory-variance patterns reached approximately 0.96.

Most importantly, the replication reproduces the main operational conclusion of the original study: when replenishment lead times become long, direct multi-step forecasts can substantially reduce order amplification relative to repeatedly extending a one-step forecast across the entire lead time.

Nevertheless, the absolute M5 forecasting errors, particularly for LSTM, and the Table 4 statistical comparisons remain different from the published values. For this reason, the results should be interpreted as evidence of a successful qualitative and partially quantitative reconstruction rather than an exact reproduction of the authors' unpublished implementation.

References

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[2] S. Makridakis, E. Spiliotis, and V. Assimakopoulos, "The M5 Accuracy Competition: Results, Findings, and Conclusions," *International Journal of Forecasting*, vol. 38, no. 4, pp. 1346–1364, 2022.